TechStore – Documentation

Structure, styling and behaviour of online shopping 2.html

1. Overview

The project is one self-contained HTML file of about 590 lines: a <style> block for all CSS, the page markup, and a <script> block at the end of <body> for behaviour. There are no external libraries, fonts or stylesheets; only the product images are fetched from Unsplash.

2. Page structure

Section	Element / selector	Purpose
Navigation	header, .logo, #search, #cart-link	Brand, search input, anchor links, cart count (#cart-count).
Hero	.hero, .shop-btn	Promo headline, text, Shop Now link to #products, side image.
Categories	.categories, .category-btn	Five pill buttons; data-category holds all / mobiles / laptops / audio / accessories.
Products	#products, #product-grid, .product-card	Eight cards; each carries data-category.
Services	.benefits	Free Delivery, Secure Shopping, Customer Support.
Footer	footer	Brand line and copyright.

Product card anatomy

```
<div class="product-card" data-category="mobiles">
  <img src="..." alt="Smartphone">
  <span class="discount">20% OFF</span>
  <h3>Smartphone</h3>
  <p class="description">...</p>
  <p class="rating">★★★★★ (4.8)</p>
  <p class="price">₹15,999 <span class="old-price">₹19,999</span></p>
  <button class="add-btn" data-name="Smartphone" data-price="15999">Add to Cart</button>
</div>
```

The button's data-name and data-price attributes feed the cart script, so a new product needs only a new card with these attributes.

3. Styling

- **Reset** – * { margin:0; padding:0; box-sizing:border-box }.
- **Colours** – primary purple #5b45e0, page background #f5f7fb, hero gradient #5541d9 → #a35be8, discount green #07864a, rating gold #f4a900, footer #211c3c.
- **Header** – Flexbox with flex-wrap so items wrap on narrow screens.
- **Product grid** – CSS Grid, repeat(4, 1fr) with 22 px gap.
- **Card hover** – lifts 5 px (translateY(-5px)) with a stronger shadow over 0.3 s.
- **Images** – fixed 175 px height with object-fit: contain.

Responsive breakpoints

Max width	Changes
950 px	Product grid becomes 2 columns.

Max width	Changes
600 px	Header content centred; search box full width; hero padding and heading reduced; hero image hidden; product grid and service strip become 1 column.

4. JavaScript behaviour

Cart. Two variables, `cartCount` and `cartTotal`, are held in memory. Each Add to Cart click reads the button's dataset, increments both, updates the header count and shows an alert such as "Smartphone added to cart! Items: 1 Cart total: ₹15,999". Totals use `toLocaleString("en-IN")` for Indian digit grouping. Clicking the cart link calls `preventDefault()` and alerts the totals.

Category filter. A click stores `currentCategory`, moves the `active` class to the clicked pill and calls `filterProducts()`.

Search. On every `input` event the lower-cased text is compared with each card's name (h3) and description. A card is shown (`display: block`) only if it matches both the search text and the current category.

Empty message. `showEmptyMessage()` adds a "No products found" paragraph when every card is hidden and removes it when any card is visible.

Functions and handlers

Name	Type	Role
Add button click	Event listener (×8)	Update count and total; show alert.
`#search` input	Event listener	Filter by text and category.
Category button click	Event listener (×5)	Set category, highlight pill, filter.
`filterProducts()`	Function	Same match logic as the search handler.
`showEmptyMessage()`	Function	Toggle the no-results message.
`#cart-link` click	Event listener	Show cart summary alert.

5. Product data

Product	Category	Price	Old price	Discount	Rating
Smartphone	Mobiles	₹15,999	₹19,999	20%	4.8
Power Laptop	Laptops	₹42,499	₹49,999	15%	4.7
Wireless Headphones	Audio	₹1,399	₹1,999	30%	4.5
Smart Watch	Accessories	₹2,249	₹2,999	25%	4.6
Wireless Earbuds	Audio	₹1,599	₹1,999	20%	4.4
Wireless Mouse	Accessories	₹899	₹999	10%	4.3
Premium Phone	Mobiles	₹21,999	₹24,999	12%	4.7
Student Laptop	Laptops	₹32,799	₹39,999	18%	4.5

6. Tested behaviour

The page was opened in a headless browser and exercised. Adding Smartphone then Wireless Mouse gave alerts of "Items: 1, ₹15,999" then "Items: 2, ₹16,898", and the cart link reported 2 items and ₹16,898 – all correct. Category and search filtering, the empty state, and the 4 / 2 / 1 column layouts at 1366, 820 and 390 px wide were also checked. See screenshots.pdf.

7. Observations and suggested improvements

- **Images blocked in testing.** Unsplash was unreachable in the test environment, so images appear as broken icons in the screenshots; they should load in a normal browser.
- **Duplicate logic.** The search handler repeats the code in `filterProducts()`; calling `filterProducts()` from the search handler would remove the duplication.
- **Declaration order.** `currentCategory` is declared with `let` after the search handler that uses it. It works because the handler runs later, but declaring it earlier is safer.
- **Cart.** Add an item list, quantities, removal, a cart panel instead of alerts, and `localStorage` persistence.
- **Accessibility.** Add `aria-label` to the search input, use real buttons or links for the cart, and describe the star ratings for screen readers.
- **Dead link.** `#cart` has no target element.
- **Trust text.** “Secure Shopping” and “Free Delivery” are placeholder claims in a demo with no checkout.